

API SECURITY RISK ANALYSIS REPORT

i. Introduction

APIs enable communication between systems but can introduce security risks if not properly secured. This report analyzes a public API to identify weaknesses and recommend improvements.

ii. API Details

Endpoint: <https://jsonplaceholder.typicode.com/users>

This API provides mock user data such as names, emails, and addresses.

iii. Methodology

The API was tested using browser, Postman, and cURL. Responses were analyzed for security weaknesses.

iv. Findings

- No Authentication
- Excessive Data Exposure
- No Rate Limiting
- No Access Control

v. Impact

Potential risks include data harvesting, phishing, unauthorized access, and service disruption.

vi. Recommendations

- Implement authentication (JWT/API keys)
- Apply rate limiting
- Reduce exposed data
- Enforce RBAC
- Monitor API logs

vii. OWASP Mapping

Broken Object Level Authorization

Broken Authentication

Excessive Data Exposure
Lack of Rate Limiting

viii. Conclusion

Proper API security controls are essential to prevent misuse and protect data.